

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

December 2011

EC 303 Audio Video Engineering

Date: 14.12.2011, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

- Q - 1 (a) Justify the choice of 625 lines for TV transmission. Why is the total number of lines kept odd in all TV system? [03]
- (b) Show that the highest modulating frequency that needs to be handled in the 625 TV system is 5 MHz [04]
- Q - 2 (a) Sketch the composite video signal waveform for at least three successive lines and indicate: (1) Extreme white level, (2) blanking level, (3) pedestal height and (4) sync pulse level. [05]
- (b) Discuss vestigial side band transmission and Show that Channel Bandwidth of Indian television standard is 7Mhz. [05]
- (c) Define (i) Brightness (ii) Hue (iii) Saturation (iv) Flicker : [04]

OR

- Q - 2 (a) Give the comparison between positive modulation and negative modulation [05]
- (b) Write a short note on "Interference suffered by TV signal" [05]
- (c) Explain Interlaced Scanning in brief [04]
- Q - 3 (a) Draw block diagram of an RF tuner and explain how incoming signals from different stations are translated to common picture IF and sound IF frequency [05]
- (b) Describe briefly the factors that influenced the choice of picture IF=38.9MHz. [05]
- (c) Explain Saw Filter with simplified diagram [04]

OR

- Q - 3 (a) Write a short note on High Definition Television (HDTV) [05]
- (b) Draw and explain block diagram of monochrome television [05]
- (c) Draw and explain block diagram of Video amplifier in B&W TV receiver [04]

SECTION - II

- Q - 4 Answer the Following [07]
- (a) Frequency response of a microphone is determined by _____ and _____ of the moving coil assembly.
- (b) _____ Microphone's directivity pattern is bi-directional.
- (c) Enclosure of a loudspeaker acts as _____

- (d) When the load resistance is lower than the source resistance _____ transformer is used for matching. (step up, step down)
- (e) Most popular microphone for use in PA system is _____
- (f) Level _____ is the remedy for those speakers who keep their mouth very near the microphone or who start shouting on the mic. (canceller, emphasizer, limiter, amplifier)
- (g) The sound would be _____ without reverberation in a concert hall.

- Q - 5 (a)** Write short note on PA system. [04]
- (b)** Calculate the efficiency of loudspeaker if power intensity of sound is 0.08 W/m^2 at 1 metre distance from the loudspeaker, when the amplifier delivers 25 watt. [05]
- (c)** Explain with the help of a net figure, how does sound energy grow and decay at a point in a reverberating room. [05]

OR

- Q - 5 (a)** If normal conversation level is 25 micro watt and we need at least 3dB more for comfortable listening in a meeting, calculate the intensity of sound required in dB over threshold of hearing, presuming 3dB of power is absorbed by the audience. Threshold of hearing is 1 Pico watt. [04]
- (b)** Give the comparison between cone type and horn type loud speakers [01]
- (c)** What are the requirements for a good auditorium for pleasant listening? Give salient features of acoustical design for an auditorium. [05]
- Q - 6 (a)** Define reverberation time. [01]
- (b)** List the Application of Multimedia. [07]
- (c)** Explain the parameters on which quality of a microphone depends. Give the value of these parameters for moving coil, ribbon, crystal, condenser and carbon microphones. [06]

OR

- Q - 6 (a)** Define hypertext, hypermedia and Multimedia with suitable example. [07]
- (b)** Reverberation is caused by _____. [01]
- (c)** The output of a directive microphone is 0.3 mV, when the output of an omni directional microphone placed under identical condition with respect to source is only 3 micro volts. Calculate directivity of the two microphones in dB. [06]
